Experiment 3

Handling missing and inappropriate data in a dataset.

Code 1

```
import numpy as np
import pandas as pd

df = pd.read_csv("Hotel_Dataset.csv")
print("Original Dataset:")
print(df)
print()

print("Duplicate rows check:")
print(df.duplicated())
print()

print("Dataset Info:")
df.info()
print()

df.drop_duplicates(inplace=True)
print("Dataset after removing duplicates:")
print(df)
print()
print("Number of rows after removing duplicates:", len(df))
print()

index = np.array(list(range(0, len(df))))
df.set_index(index, inplace=True)
print("Dataset after resetting index:")
print(df)
print()

df.drop(['Age_Group.1'], axis=1, inplace=True)
print("Dataset after dropping redundant column:")
print(df)
print()

df['CustomerID'] = df['CustomerID'].mask(df['CustomerID'] < 0)
df['Bill'] = df['Bill'].mask(df['Bill'] < 0)
df['EstimatedSalary'] = df['EstimatedSalary'].mask(df['EstimatedSalary'] < 0)
df['NoOfPax'] = df['NoOfPax'].mask((df['NoOfPax'] < 1) | (df['NoOfPax'] > 20))
print("Dataset after converting inappropriate values to NaN:")
print(df)
print()

print("Unique Age_Group values:", df.Age_Group.unique())
print("Unique Hotel values:", df.Hotel.unique())
print("Unique FoodPreference values:", df.FoodPreference.unique())
print()

df['Hotel'] = df.Hotel.replace(['Ibys'], 'Ibis')
df['FoodPreference'] = df.FoodPreference.replace(['Vegetarian', 'veg'], 'Veg')
df['FoodPreference'] = df.FoodPreference.replace(['non-Veg'], 'Non-Veg')
print("Unique Hotel values after cleaning:", df.Hotel.unique())
print("Unique FoodPreference values after cleaning:", df.FoodPreference.unique())
print()




df['EstimatedSalary'] = df.EstimatedSalary.fillna(round(df.EstimatedSalary.mean()))
df['NoOfPax'] = df.NoOfPax.fillna(round(df.NoOfPax.median()))
df['Rating(1-5)'] = df['Rating(1-5)'].fillna(round(df['Rating(1-5)'].median()))
df['Bill'] = df.Bill.fillna(round(df.Bill.mean()))
print("Final Cleaned Dataset:")
print(df)
```

Output

```
Original Dataset:
    CustomerID Age_Group  Rating(1-5)      Hotel FoodPreference  Bill  \
0            1     20-25            4       Ibis            veg  1300
1            2     30-35            5  LemonTree        Non-Veg  2000
2            3     25-30            6     RedFox            Veg  1322
3            4     20-25           -1  LemonTree            Veg  1234
4            5       35+            3       Ibis     Vegetarian   989
```

```
5            6       35+            3       Ibys        Non-Veg  1909
6            7       35+            4     RedFox     Vegetarian  1000
7            8     20-25            7  LemonTree            Veg  2999
8            9     25-30            2       Ibis        Non-Veg  3456
9            9     25-30            2       Ibis        Non-Veg  3456
10          10     30-35            5     RedFox        non-Veg -6755

    NoOfPax  EstimatedSalary Age_Group.1
0         2            40000       20-25
1         3            59000       30-35
2         2            30000       25-30
3         2           120000       20-25
4         2            45000         35+
5         2           122220         35+
6        -1            21122         35+
7       -10           345673       20-25
8         3           -99999       25-30
9         3           -99999       25-30
10        4            87777       30-35

Duplicate rows check:
0     False
1     False
2     False
3     False
4     False
5     False
6     False
7     False
8     False
9      True
10    False
dtype: bool

Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11 entries, 0 to 10
Data columns (total 9 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   CustomerID       11 non-null     int64
 1   Age_Group        11 non-null     object
 2   Rating(1-5)      11 non-null     int64
 3   Hotel            11 non-null     object
 4   FoodPreference   11 non-null     object
 5   Bill             11 non-null     int64
 6   NoOfPax          11 non-null     int64
 7   EstimatedSalary  11 non-null     int64
 8   Age_Group.1      11 non-null     object
dtypes: int64(5), object(4)
memory usage: 924.0+ bytes

Dataset after removing duplicates:
    CustomerID Age_Group  Rating(1-5)      Hotel FoodPreference  Bill  \
0            1     20-25            4       Ibis            veg  1300
1            2     30-35            5  LemonTree        Non-Veg  2000
2            3     25-30            6     RedFox            Veg  1322
3            4     20-25           -1  LemonTree            Veg  1234
4            5       35+            3       Ibis     Vegetarian   989
5            6       35+            3       Ibys        Non-Veg  1909
6            7       35+            4     RedFox     Vegetarian  1000
7            8     20-25            7  LemonTree            Veg  2999
8            9     25-30            2       Ibis        Non-Veg  3456
10          10     30-35            5     RedFox        non-Veg -6755

    NoOfPax  EstimatedSalary Age_Group.1
0         2            40000       20-25
1         3            59000       30-35
2         2            30000       25-30
3         2           120000       20-25
4         2            45000         35+
5         2           122220         35+
6        -1            21122         35+
7       -10           345673       20-25
8         3           -99999       25-30
10        4            87777       30-35

Number of rows after removing duplicates: 10

Dataset after resetting index:
   CustomerID Age_Group  Rating(1-5)      Hotel FoodPreference  Bill  NoOfPax  \
0           1     20-25            4       Ibis            veg  1300        2
1           2     30-35            5  LemonTree        Non-Veg  2000        3
2           3     25-30            6     RedFox            Veg  1322        2
```

```
3           4     20-25           -1  LemonTree            Veg  1234        2
4           5       35+            3       Ibis     Vegetarian   989        2
5           6       35+            3       Ibys        Non-Veg  1909        2
6           7       35+            4     RedFox     Vegetarian  1000       -1
7           8     20-25            7  LemonTree            Veg  2999      -10
8           9     25-30            2       Ibis        Non-Veg  3456        3
9          10     30-35            5     RedFox        non-Veg -6755        4

   EstimatedSalary Age_Group.1
0            40000       20-25
1            59000       30-35
2            30000       25-30
3           120000       20-25
4            45000         35+
5           122220         35+
6            21122         35+
7           345673       20-25
8           -99999       25-30
9            87777       30-35

Dataset after dropping redundant column:
   CustomerID Age_Group  Rating(1-5)      Hotel FoodPreference  Bill  NoOfPax  \
0           1     20-25            4       Ibis            veg  1300        2
1           2     30-35            5  LemonTree        Non-Veg  2000        3
2           3     25-30            6     RedFox            Veg  1322        2
3           4     20-25           -1  LemonTree            Veg  1234        2
4           5       35+            3       Ibis     Vegetarian   989        2
5           6       35+            3       Ibys        Non-Veg  1909        2
6           7       35+            4     RedFox     Vegetarian  1000       -1
7           8     20-25            7  LemonTree            Veg  2999      -10
8           9     25-30            2       Ibis        Non-Veg  3456        3
9          10     30-35            5     RedFox        non-Veg -6755        4

   EstimatedSalary
0            40000
1            59000
2            30000
3           120000
4            45000
5           122220
6            21122
7           345673
8           -99999
9            87777

Dataset after converting inappropriate values to NaN:
   CustomerID Age_Group  Rating(1-5)      Hotel FoodPreference    Bill  \
0           1     20-25            4       Ibis            veg  1300.0
1           2     30-35            5  LemonTree        Non-Veg  2000.0
2           3     25-30            6     RedFox            Veg  1322.0
3           4     20-25           -1  LemonTree            Veg  1234.0
4           5       35+            3       Ibis     Vegetarian   989.0
5           6       35+            3       Ibys        Non-Veg  1909.0
6           7       35+            4     RedFox     Vegetarian  1000.0
7           8     20-25            7  LemonTree            Veg  2999.0
8           9     25-30            2       Ibis        Non-Veg  3456.0
9          10     30-35            5     RedFox        non-Veg     NaN

   NoOfPax  EstimatedSalary
0      2.0          40000.0
1      3.0          59000.0
2      2.0          30000.0
3      2.0         120000.0
4      2.0          45000.0
5      2.0         122220.0
6      NaN          21122.0
7      NaN         345673.0
8      3.0              NaN
9      4.0          87777.0

Unique Age_Group values: ['20-25' '30-35' '25-30' '35+']
Unique Hotel values: ['Ibis' 'LemonTree' 'RedFox' 'Ibys']
Unique FoodPreference values: ['veg' 'Non-Veg' 'Veg' 'Vegetarian' 'non-Veg']

Unique Hotel values after cleaning: ['Ibis' 'LemonTree' 'RedFox']
Unique FoodPreference values after cleaning: ['Veg' 'Non-Veg']

Final Cleaned Dataset:
   CustomerID Age_Group  Rating(1-5)      Hotel FoodPreference    Bill  \
0           1     20-25            4       Ibis            Veg  1300.0
1           2     30-35            5  LemonTree        Non-Veg  2000.0
2           3     25-30            6     RedFox            Veg  1322.0
3           4     20-25           -1  LemonTree            Veg  1234.0
4           5       35+            3       Ibis            Veg   989.0
```

5	6	35+	3	Ibis	Non-Veg	1909.0
6	7	35+	4	RedFox	Veg	1000.0
7	8	20-25	7	LemonTree	Veg	2999.0
8	9	25-30	2	Ibis	Non-Veg	3456.0
9	10	30-35	5	RedFox	Non-Veg	1801.0

	NoOfPax	EstimatedSalary
0	2.0	40000.0
1	3.0	59000.0
2	2.0	30000.0
3	2.0	120000.0
4	2.0	45000.0
5	2.0	122220.0
6	2.0	21122.0
7	2.0	345673.0
8	3.0	96755.0
9	4.0	87777.0